NEXAFRICA MACHINE LEARNING INTERNSHIP

Customer Churn Prediction

Finding the customers who are about to leave, before they do

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Losing a customer costs more than it seems

Replacing is expensive

Winning a new customer takes marketing spend, sales time and onboarding effort.

Future revenue is lost

The company loses everything that customer would have paid over the rest of the relationship.

The goal of the model

Give every customer a churn probability, so retention offers go to the people most likely to leave.

The data: 7,043 telecom customers

7,043

customers, one row each

21

columns: demographics, services, contract and billing

26.5%

of customers churned, 73.5% stayed

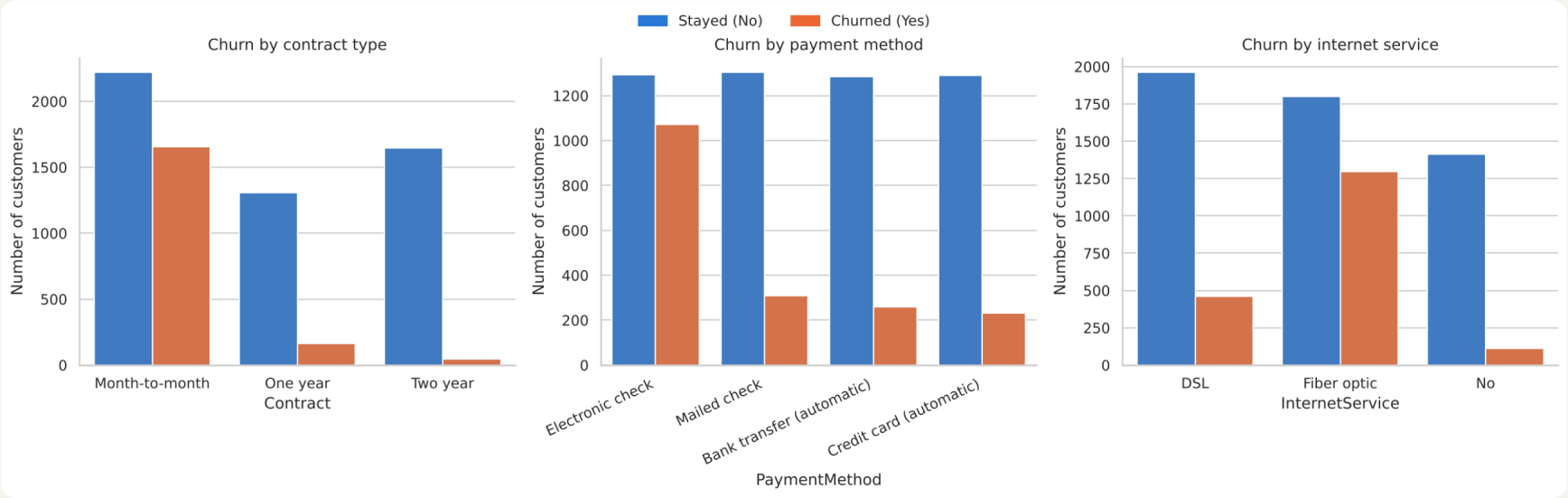
Target: Churn (Yes or No). With this split, a model that always says "stay" would be right 73% of the time and catch no one, so accuracy alone cannot be trusted.

Data preparation: one hidden problem

Check	What we found	What we did
Missing values	TotalCharges looked complete, but 11 rows held a blank space. All were new customers with tenure 0.	Converted to numbers and filled with 0
Duplicates	None found	Nothing needed
Data types	TotalCharges was stored as text	Now a numeric column
Categories	No typos or spacing errors	Nothing needed

Why it matters: a standard null check reported zero missing values, so the problem stayed invisible until the column type was questioned.

Churn is concentrated in a few groups



42.7%
churn on month-to-month contracts, versus 2.8% on two-year

45.3%
churn among electronic check payers

41.9%
churn on fiber optic; churners' median tenure is 10 months versus 38

Three new features from existing columns

Feature	How it was built	Why it helps
NumServices	Count of add-on services each customer uses	Captures how deeply a customer is tied to the company
TenureGroup	Tenure in months grouped into bands	Lets the model see that risk changes with customer age
HasInternetService	Yes or no flag from the internet service column	Separates phone-only customers from the rest

Categorical columns were one-hot encoded and numeric columns scaled, so every model starts from the same 80/20 stratified split.

Three models, three stages each

Logistic Regression

Simple and easy to explain. Each feature gets a weight that shows its push toward leaving or staying.

Decision Tree

Asks a chain of yes or no questions. Readable, but it can overfit without limits on depth.

Random Forest

Many trees voting together. Usually stronger and steadier, but harder to explain.

Each was tested three ways: a plain baseline, with class weights to counter the 73/26 imbalance (SMOTE was also tried), and tuned with cross-validated search.

After tuning, the models are close

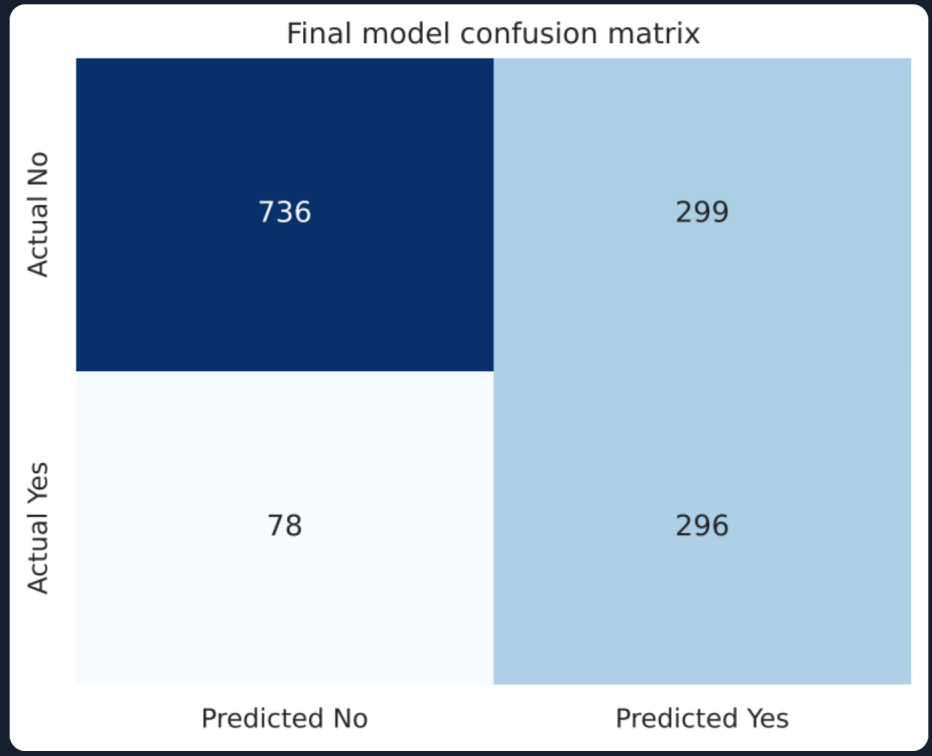
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic, baseline	0.800	0.655	0.519	0.579	0.842
Logistic, class-weighted	0.732	0.498	0.791	0.611	0.842
Decision Tree, tuned	0.742	0.509	0.810	0.625	0.838
Random Forest, tuned	0.753	0.523	0.781	0.627	0.842

Accuracy drops when recall rises, and that is the right trade: missing a churner costs more than a wasted offer. With scores this close, the choice rests on stable recall in cross-validation (0.80 for logistic versus 0.76 for the forest) and on how easily the model can be explained.

FINAL MODEL: CLASS-WEIGHTED LOGISTIC REGRESSION

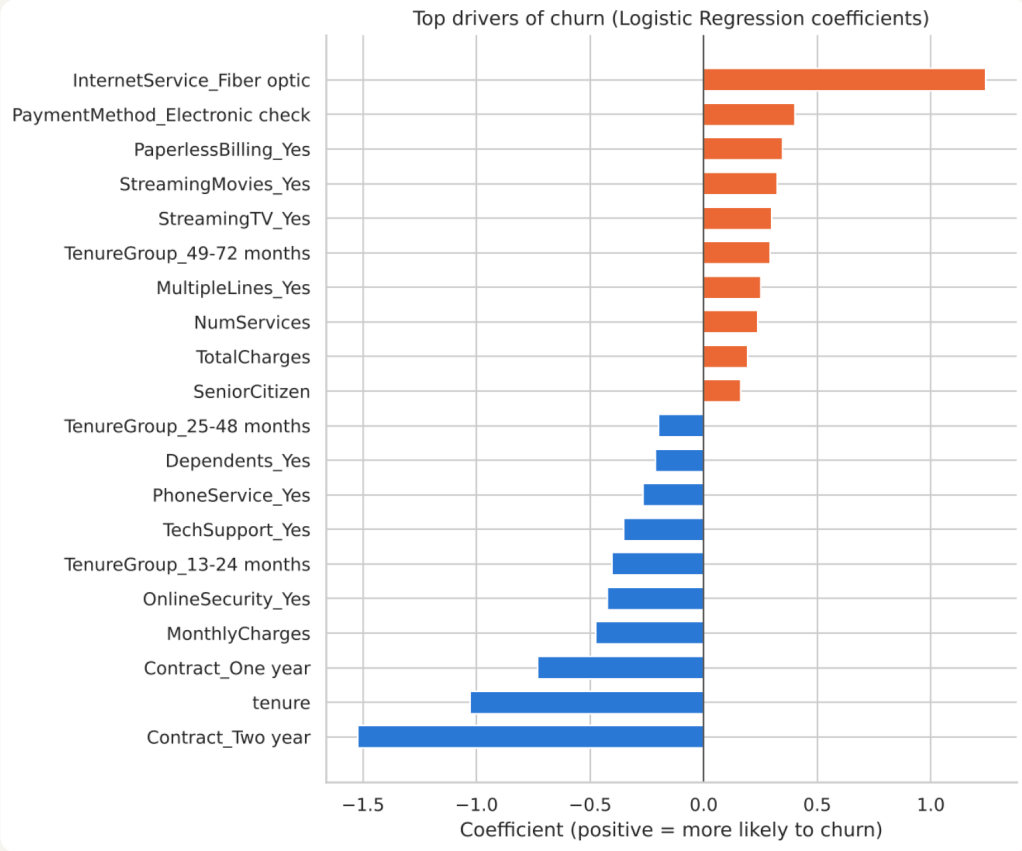
Recall rose from 52% to 79%

Of 374 customers who churned in the test set, the model caught 296 and missed 78. It raised 299 false alarms among customers who stayed.



What the model pays attention to

- Contract length and fiber optic internet carry the largest weights toward churn.
- Longer tenure and two-year contracts push strongly toward staying.
- Tenure, total charges and monthly charges overlap, so their individual weights should be read with care.
- These are associations, not proof of cause.



Six actions for the retention team

Move customers to longer contracts
Churn is 42.7% month-to-month and 2.8% on two-year plans.

Encourage automatic payment
Electronic check payers churn at 45.3%.

Review fiber optic quality and price
41.9% of fiber customers leave.

Invest in the first year
Churners' median tenure is 10 months, so onboarding matters.

Bundle support and security
Customers with these add-ons churn about 15%, versus 42% without.

Spend by churn probability
Rank customers with the model and offer to the highest risks first.

Limits and next steps

Limits to keep in mind

Precision is about 50%, so half of the flagged customers would have stayed. Findings show association, not cause. Scaling was applied before the split, which can leak a little information.

Next steps

Tune the decision threshold to the cost of an offer. Try gradient boosting such as XGBoost. Pilot the retention actions on a small group and measure the result.

Thank you. Questions?